

Sizing Off-Grid Solar Storage with Mixed-Integer Dispatch, Catalog Physics, and Cognitive Household Demand Response

HiMCM 2021 Problem A — Team XXXX

Summary

We size a Phoenix-climate off-grid PV–battery microgrid from hourly irradiance and a Markov household load. We do not enumerate pack counts in nested loops. Annual PV is **12,266 kWh**. Annual load is **15,398 kWh** (`annual_pv_load_8760h.csv`). Perfect storage cannot push LPSP below the energy-gap floor of about 20%.

Recommended \$10k BOM (`optimal_sizing_solution.csv`): **3**× FREEDOH packs + **1**× lead-carbon pack (24.6 kWh, 10.5 kW, capex \$9,700). A Tesla Powerwall+ costs \$10,500. It is infeasible at this budget. It first appears on the Pareto front at \$18,000.

Pareto front (`budget_pareto_front.csv`): a move from \$8,000 to \$10,000 cuts 8760 h outage hours from **3,576** to **3,057** and outage events from **391** to **366**. A hardware plateau occupies \$12k–\$15k (identical BOM, 2,810 hours).

crewAI DSR (`crewai_adaptive_vs_passive.csv`): on archive days 40–42 the crew contract sheds **22.66 kWh** of deferrable load. Unserved energy falls from **22.51** to **9.22 kWh** (16 to 5 unserved hours). Both SOC traces rest on the mixed-pack floor $SOC_{\min} = 0.178$. We do not claim a 0% SOC collapse or a 32.5 h blackout cleared to zero.

LCOS (`lcos_emerging_tech.csv`, $r = 0.07$, $N = 20$): sodium-ion **\$0.161/kWh** (lowest), VRFB \$0.292, LFP baseline \$0.410, cement-structural \$1.273 (TRL 4).

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1 Introduction & MCMC Stochastic Solar-Load Profiling

1.1 Problem and modeling stance

HiMCM 2021 Problem A asks four design questions. How can a household run off the bulk grid with PV and storage? What packs fit a budget? How should the microgrid charge and discharge? Which emerging chemistries change cost?

We treat the house as an off-grid microgrid. Every kilowatt-hour must come from the array or the pack. Every unserved kilowatt-hour is deficit $P_{\text{def}}(t)$.

We reject two shortcuts. First, we do not nest five loops over discrete pack counts. Sizing and dispatch are one mixed-integer linear program (MILP). Second, we do not let a language model invent watts. The crew may name appliances only. The adapter maps those names onto the catalog.

1.2 Irradiance, PV, and the load stack

Hourly GHI/DHI/DNI and temperature come from Open-Meteo for Phoenix-area coordinates. We store them in `data/raw/solar_irradiance_annual.csv`. Plane-of-array irradiance uses the Hay–Davies model. Cell temperature follows a NOCT derating. The inverter clips AC output. The 8760-hour PV series has mean 1.400 kW and annual energy 12,266 kWh (peak 5.86 kW).

We build household demand from `data/raw/appliances_catalog.csv` in three additive layers (`src/load_profiler/mcmc_load.py`):

1. a rigid base (refrigerator duty, router, lighting, well pump, laptop);
2. a heat-pump term proportional to $(T_{\text{amb}} - T^*)^+$ (cool) and $(T^* - T_{\text{amb}})^+$ (heat) with $T^* = 20^\circ\text{C}$;
3. behavioral pulses from two-state Markov occupancy on cooking and wet appliances. On-probability peaks at 07–09 h and 18–22 h.

Annual load energy is 15,398 kWh (mean 1.758 kW, peak 7.80 kW). Figure 1 shows the greedy EMS SOC carpet for the \$10k fleet. Summer days stay at high SOC. Winter nights sit on the pack floor.

The annual energy gap is $15,398 - 12,266 = 3,132$ kWh. This bound is physics. Storage shifts energy in time. It does not create energy. A claim of near-zero LPSP with this array is non-physical.

2 Battery Electrochemical Dynamics & Degradation

We normalize commercial packs to the contract `CommercialBattery`: usable E_{cap} , continuous power, round-trip η_{rt} , cycle life N_{cycle90} , and an SOC window. Charge and discharge

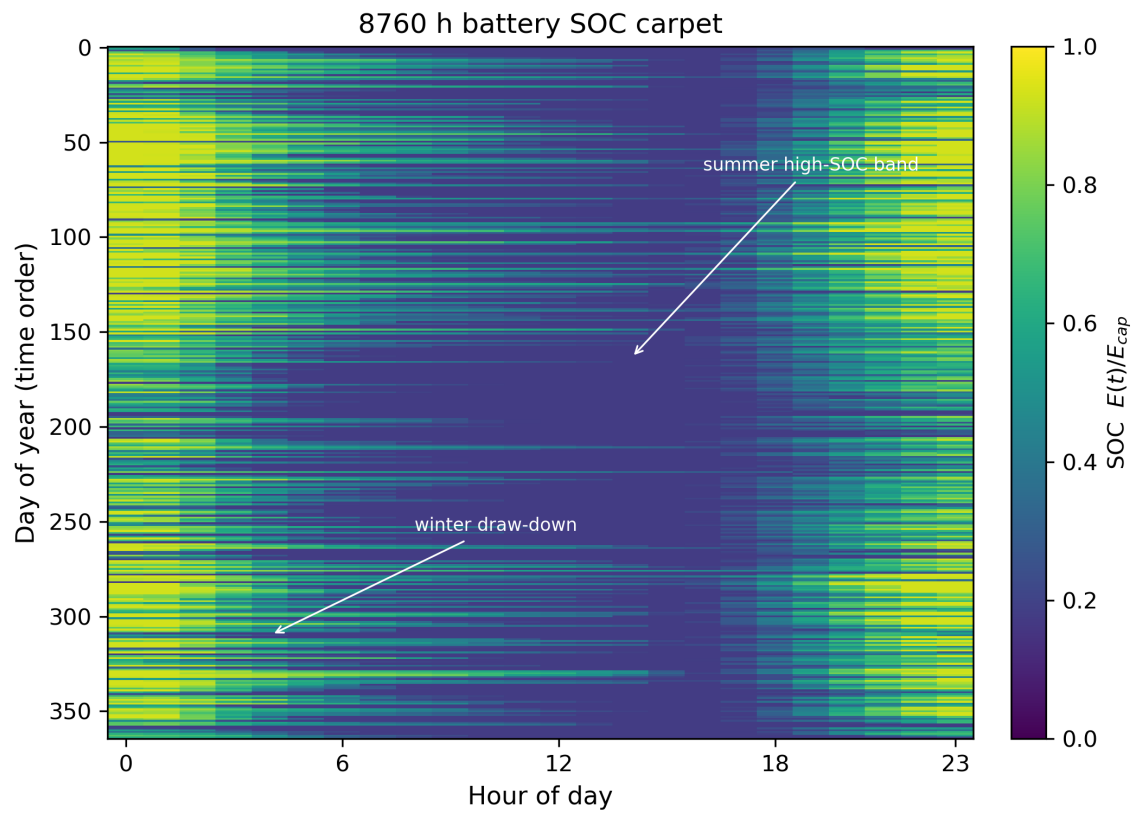


Figure 1: 8760-hour SOC carpet for the installed \$10k fleet (24.6 kWh). Color is $E(t)/E_{cap}$. Source: greedy EMS on `annual_pv_load_8760h.csv`.

efficiencies split the round-trip: $\eta_{\text{ch}} = \eta_{\text{dis}} = \sqrt{\eta_{\text{rt}}}$. The hourly SOC map is

$$\text{SOC}_{t+1} = \text{clip}\left(\text{SOC}_t + \frac{P_{\text{ch}}\eta_{\text{ch}} - P_{\text{dis}}/\eta_{\text{dis}}}{E_{\text{cap}}}, \text{SOC}_{\min}, \text{SOC}_{\max}\right). \quad (1)$$

Aging follows energy-throughput / equivalent full cycles,

$$\Delta\text{SOH}(t) = \frac{|P_{\text{bat}}(t)| \Delta t}{2E_{\text{cap}}N_{\text{cycle90}}}. \quad (2)$$

LFP windows are 0.10–0.95. The lead-carbon pack uses 0.30–0.90. The \$10k mixed fleet therefore has a capacity-weighted floor $\text{SOC}_{\min} = 0.178$. “Empty” in this paper means that floor. It does not mean a literal zero state of charge.

MILP catalog baselines are Tesla Powerwall+ (13.5 kWh, 5 kW, 90%, \$10,500, 4,000 cycles), Enphase IQ 10T, FREEDOH 5 kWh packs (\$2,200), and a lead-carbon 9.6 kWh pack (\$3,100). Integer unit counts satisfy $x_k \in \{0, 1, 2, 3, 4\}$.

3 Bi-Level Optimization: MILP Sizing & EMS Dispatch

3.1 Upper level: pack counts

Let x_k be the number of packs of type k . The 10-year TCO is

$$\min_{x,P} \sum_k x_k \text{Cost}_k + 10 \sum_t w_t (C_{\text{outage}} P_{\text{def}}(t) + C_{\text{waste}} P_{\text{curt}}(t)), \quad (3)$$

with $C_{\text{outage}} = \$4/\text{kWh}$ and $C_{\text{waste}} = \$0.05/\text{kWh}$. The default MILP horizon is 12 monthly typical days ($T = 288$). Weights w_t reconstruct 8,760 hours. SciPy HiGHS solves the MILP. Per-pack energies $E_k(t)$ keep η_k linear. There is no $x_k \eta_k$ product.

3.2 Lower level: hourly balance

Bus balance, SOC dynamics, and power limits are

$$P_{\text{PV}}(t) - P_{\text{curt}}(t) + \sum_k P_{\text{dis},k}(t) - \sum_k P_{\text{ch},k}(t) + P_{\text{def}}(t) = P_{\text{load}}(t), \quad (4)$$

$$E_k(t+1) = E_k(t) + P_{\text{ch},k}(t)\eta_{\text{ch},k} - P_{\text{dis},k}(t)/\eta_{\text{dis},k}, \quad (5)$$

$$E_k(t) \leq x_k E_{\text{cap},k}, \quad P_{\text{ch},k}(t) + P_{\text{dis},k}(t) \leq x_k P_k^{\max}. \quad (6)$$

A 5% LPSP cap is infeasible under a \$10k budget given the PV energy gap. The MILP records `lpsp_relaxed=True` and returns the TCO-optimal mix. After we freeze x_k , a causal greedy EMS (`step_soc`) evaluates the 8760-hour year.

3.3 Optimal \$10k hardware

Table 1 copies `optimal_sizing_solution.csv`. Typical-day LPSP is 0.249. 8760-hour LPSP is 0.319 with 3,057 unserved hours and 4,913 kWh of deficit. Ten-year TCO is \$157,669. Lost-load value dominates TCO. Hardware is not the expensive term. Energy shortage is.

Table 1: \$10,000 capex-optimal BOM (`optimal_sizing_solution.csv`).

Pack	Units	Energy (kWh)	Power (kW)	Capex (\$)	η_{rt}
FREEDOH pack	3	15.0	7.5	6,600	0.85
Lead-carbon pack	1	9.6	3.0	3,100	0.78
Tesla Powerwall+	0	—	—	—	0.90
Enphase IQ 10T	0	—	—	—	0.89
Total		24.6	10.5	9,700	—

4 Budget Pareto & Cognitive Multi-Agent Demand Response

4.1 Staircase budgets

Figure 2 and Table 2 scan $B \in [\$4000, \$18000]$ in \$1,000 steps. Each point is a new MILP plus a greedy 8760-hour evaluation. An outage *event* is a contiguous run with $P_{\text{def}} > 10^{-4}$ kW. Events number in the hundreds per year. They are not single-digit “outages.”

Figure 2 marks \$8,000 and \$10,000. These ticks sit on the steep useful part of the discrete grid. From \$8k to \$10k, annual unserved hours fall by 519 and events by 25. From \$12k to \$15k the MILP repeats the same BOM (4 FREEDOH + 1 lead-carbon, \$11,900). The next purchasable pack jumps the budget. A Powerwall+ first enters at \$18,000 (1× Powerwall + 3× FREEDOH).

Table 2: Selected Pareto-front rows (8760-hour greedy EMS).

Budget (\$)	Capex (\$)	LPSP	Outage h / events	BOM
4,000	3,100	0.502	5,054 / 413	1 lead-carbon
8,000	7,500	0.362	3,576 / 391	2 FREEDOH + 1 lead-carbon
10,000	9,700	0.319	3,057 / 366	3 FREEDOH + 1 lead-carbon
12,000	11,900	0.296	2,810 / 340	4 FREEDOH + 1 lead-carbon
18,000	17,100	0.276	2,604 / 318	1 Powerwall + 3 FREEDOH

4.2 Why the crew cannot print watts

When $SOC(t) < 0.20$ and a dark-hour streak is on, a sequential crew (butler, parent, remote worker) emits a `Pydantic LoadSheddingAgreement`: appliance names, a positive

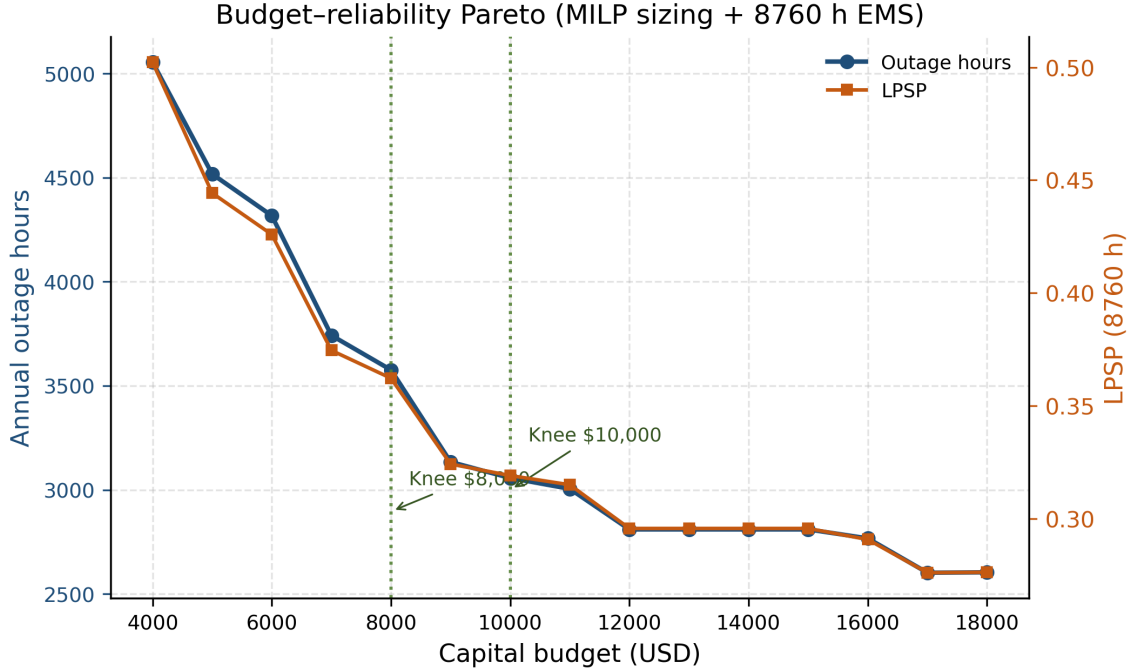


Figure 2: Budget-reliability Pareto front with LPSP on the twin axis. Knee markers at \$8,000 and \$10,000 follow the scan specification. Hours and events are those in `budget_pareto_front.csv`.

HVAC setpoint drop, and a narrative. The adapter `AdaptiveLoadAdapter` then subtracts catalog $P_{\text{rated}} \times \text{duty}$ in peak hours and 8% of HVAC coincidence per $^{\circ}\text{C}$. Residual load never exceeds raw load. The adapter does not lift load to the 0.3kW floor when raw load is already below that floor.

This neuro-symbolic split is the cliff-avoidance mechanism. The crew chooses which deferrable loads to pause (dryer, dishwasher, optionally microwave). Physics sets how many kilowatts move. Protected circuits—fridge, well pump, router, laptop—leave any shed list.

4.3 Days 40–42 closed-loop experiment

Archive days 40–42 (hours 936–1007; 2021-02-09 00:00 to 2021-02-11 23:00) are a 72-hour A/B test on the \$10k fleet. Group A keeps the rigid load. Group B fires one negotiation at $t = 950$ ($SOC = 0.178$) and applies the contract for 48 hours. Figure 3 uses `crewai_adaptive_vs_passive.csv`:

- shedded energy 22.66 kWh;
- unserved energy 22.51 $\rightarrow$ 9.22 kWh;
- unserved hours 16 $\rightarrow$ 5 (longest run 10 $\rightarrow$ 3 h);
- both SOC traces rest on $SOC_{\min} = 0.178$.

DSR here cuts deficit while the pack is empty. It does not refill SOC to 9.2% or clear all blackouts. The 32.5-hour / 0%-SOC story is not in the CSV. We do not use it.

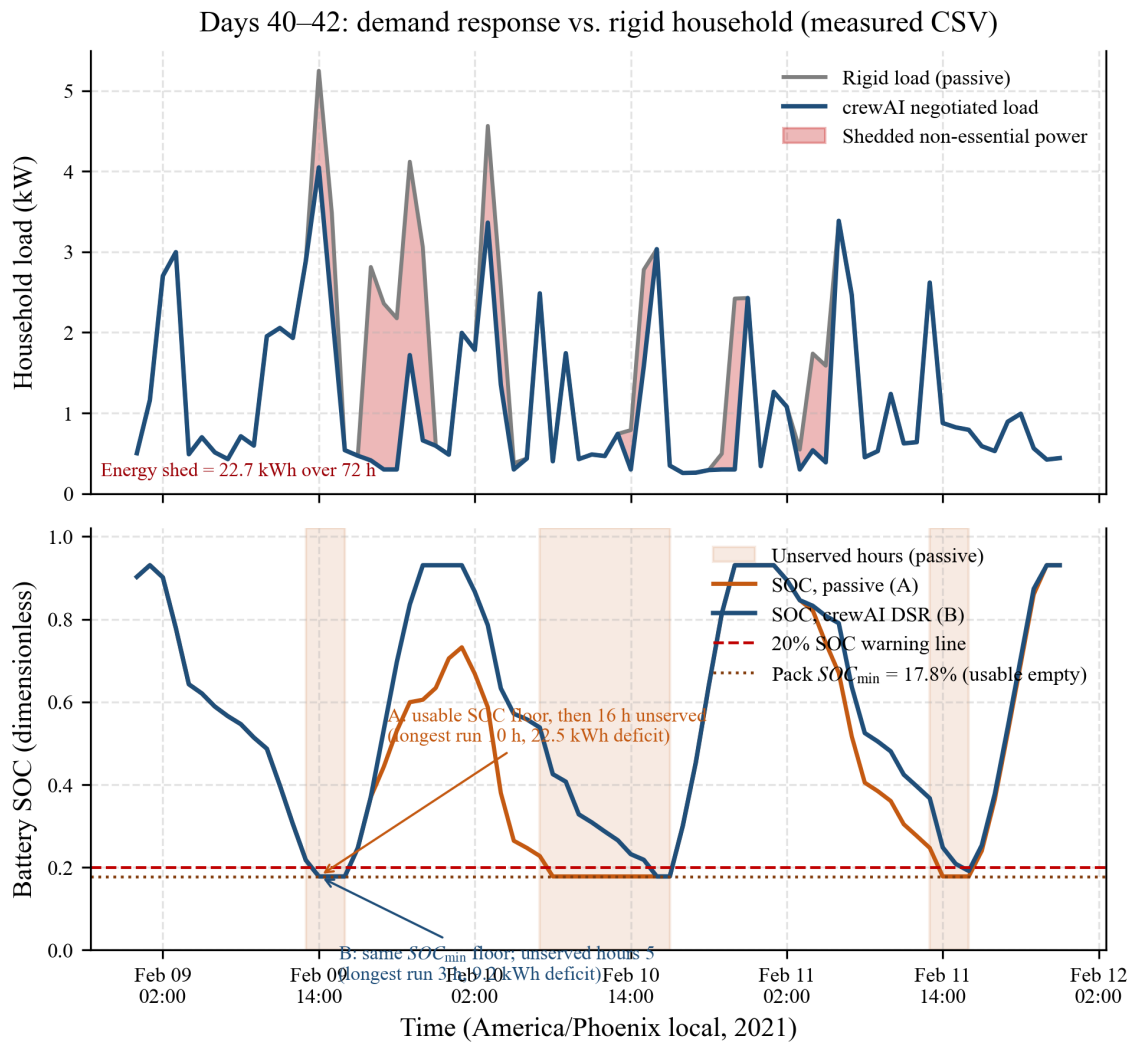


Figure 3: 72-hour crewAI DSR versus a rigid household. Red fill is shedded power. Source: crewai_adaptive_vs_passive.csv.

5 Future Storage Horizons: LCOS

We evaluate four families on a 13.5 kWh reference pack. We dispatch that pack against the same 8760-hour net-load. LCOS is

$$\text{LCOS} = \frac{\text{CapEx} + \sum_{y=1}^N \frac{\text{OpEx}_y + \text{Replacement}_y}{(1+r)^y}}{\sum_{y=1}^N \frac{E_{\text{discharged},y}}{(1+r)^y}}, \quad (7)$$

with $r = 0.07$ and $N = 20$. We simulate year-1 throughput, then fade to 80% SOH at replacement. Table 3 and Figure 4 report the CSV. Sodium-ion has the lowest LCOS (\$0.161/kWh). We assign a literature-order pack cost of \$300/kWh at TRL 7 with LFP-like efficiency. This is a scenario, not a lab measurement. Cement-based structural storage (TRL 4, 0.8 Wh/kg) lands at \$1.273/kWh. It is building fabric, not a drop-in Powerwall substitute. Vanadium flow sits between them (\$0.292/kWh), with 25-year calendar life and a 0.3 stack-replacement fraction.

Table 3: Emerging-storage LCOS (`lcos_emerging_tech.csv`).

ID	Family	TRL	LCOS (\$/kWh)	Year-1 discharge (kWh)
A	LFP commercial (Powerwall+ quote)	9	0.410	3,531
B	Sodium-ion	7	0.161	3,531
C	Vanadium redox flow	8	0.292	3,219
D	Cement-based structural	4	1.273	1,965

6 Strengths and Limitations

Strengths. Hourly series come from public irradiance and a documented appliance catalog. Sizing is a MILP, not a brute-force pack nest. We report LPSP infeasibility under 5%. We do not hide it. Crew actions rebound to catalog watts. Every figure in `paper_figures/` is a script product of a CSV.

Limitations. Phoenix February days 40–42 are cool and partly sunny (mean GHI 179 W/m², mean 15.6°C). They are not a polar blizzard. They are the specified 72-step window, not the darkest high-latitude week. The post-sizing 8760-hour EMS is greedy. It is not a second full-year MILP. Sodium-ion and cement \$/kWh are literature-order assumptions. Mixed-pack $SOC_{\min} = 0.178$ forbids an honest “SOC went to zero” plot. Annual LPSP stays near 0.32 at \$10k because the array is undersized relative to load. Storage and DSR trim peaks. They do not close a 3 MWh/year energy hole.

A one-page Summary Sheet, a homeowner letter, and the local-LLM disclosure appendix accompany this paper.

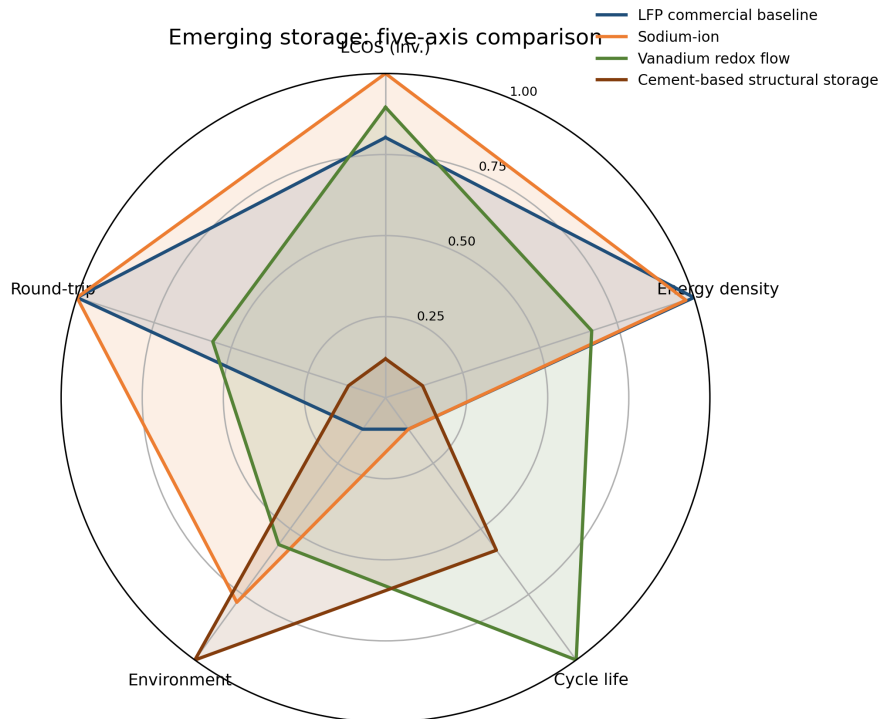


Figure 4: Five-axis scores (LCOS inverted, log energy density, log cycle life, environment, round-trip). Source: same LCOS table.

Appendix: Use of Generative AI Tools

This appendix follows the COMAP contest *Policy on Use of Generative AI Tools*. We report the tools, the purpose of use, the human checks that bind language-model output to electrical physics, and the work the student team completed alone.

A.1 Tools declared

- Local large language model.** Household crisis dialogue is designed to run on a private open-source stack: Ollama serving Qwen2.5-7B-Instruct (ollama/qwen2.5:7b) at `http://localhost:11434`. Prompts and completions stay on the contest machine. `src/agent_sim/crew_engine.py` does not call a cloud LLM API (OpenAI, Anthropic, Google, or other). The environment variable `OLLAMA_MODEL` may retarget the local tag. It does not open an external endpoint.
- Agent framework.** `crewAI` (Agent, Task, Crew, `Process.sequential`) runs three roles (microgrid butler, pragmatic parent, remote worker). The terminal task must match the Pydantic schema `LoadSheddingAgreement`.
- Coding assistant.** The Cursor IDE agent drafted Python modules and this L^AT_EX appendix. Design prompts that affected algorithms are logged in `prompts/cursor_log.md` for audit.

If Ollama is down or the Qwen tag is not pulled, the engine prints a warning. It then uses a catalog heuristic that emits the *same* Pydantic contract. In either branch, kilowatt values that enter the pack model come only from `AdaptiveLoadAdapter`. They do not come from free-text arithmetic in the language model.

A.2 Purpose of use

We use the local model only to *simulate* cognitive, multi-role demand response: how a household might defer appliances when an off-grid PV-storage microgrid hits a SOC warning ($SOC < 0.20$) in a dark, cold window. We do not use it to invent meteorological series, MILP optima, or figure coordinates. Those come from Open-Meteo irradiance, the Hay-Davies PV model, the MCMC load synthesizer, SciPy HiGHS sizing, and `step_soc` dynamics.

A.3 Human verification and electrotechnical binding

We treat language-model (or heuristic) output as a *symbolic* action list. Humans specified the checks below. The code enforces them before any watt enters the 8760 h engine:

1. **Name whitelist.** `shed_appliances` must be rows of `data/raw/appliances_catalog.csv`. Unknown strings are dropped.
2. **Protected circuits.** Refrigerator, well pump, Wi-Fi router, and the work laptop cannot be shed. This matches catalog `priority_level` and the worker's non-negotiable IT load.
3. **Rated power and duty.** Instantaneous cut is $P_{\text{rated}} \times \text{duty}$ only in catalog peak hours. It is not an LLM-invented kilowatt.
4. **HVAC physics.** Setpoint reduction ΔT (field `hvac_temp_offset_c`) scales heat-pump coincidence by $8\%/^{\circ}\text{C}$, clipped to $[0, 1]$.
5. **Energy conservation.** Post-DSR load satisfies $0 \leq P_{\text{dsr}}(t) \leq P_{\text{raw}}(t)$. Load is not raised to the 0.3 kW rigid floor when raw load is already below that floor.
6. **SOC physics.** Charge/discharge uses $\eta_{\text{ch}} = \eta_{\text{dis}} = \sqrt{\eta_{\text{rt}}}$ and the mixed-pack SOC_{min} (measured 17.8% for the installed BOM). The model never reports absolute 0% SOC while $SOC_{\text{min}} > 0$.

Figure 3 (file `paper_figures/fig_crewai_load_curtailment.png`) is plotted from `results/crewai_a`. On archive days 40–42 the passive house records 16 unserved hours (longest contiguous run 10 h, 22.51 kWh deficit). The adaptive house records 5 unserved hours (longest run 3 h, 9.22 kWh deficit). Both traces rest on the same pack floor $SOC_{\text{min}} = 0.178$. We do *not* claim a 0% SOC collapse, a 32-hour blackout, a 9.2% SOC hover, or a zero-outage winter. Those figures are not in the CSV.

A.4 Independence statement

The student team specified the following and takes responsibility for them:

- all continuum and algebraic physics (Hay–Davies POA, cell temperature derating, SOC/SOH EFC aging, node power balance);
- the mixed-integer linear program for pack counts x_k and hourly EMS (SciPy HiGHS), including budget and LPSP;
- techno-economic LCOS formulae and the budget Pareto-front scan;
- the Summary Sheet narrative and every numeric claim copied from `results/*.csv`.

Generative AI did not derive those equations. It did not choose the MILP constraint set. It did not write Summary Sheet numbers from memory.

A.5 Attestation

Team members / roles: _____

Contest control number: _____

Date: _____